

Hong Tien Nguyen

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EDUCATION

University of Southern California, Los Angeles, CA

Sep 2021 - Expected Dec 2027

Ph.D., Electrical and Computer Engineering — Advisors: Shrikanth Narayanan

- Research Interests: Human-centered Video Understanding, Temporal Dynamics in the context of Medical Imaging, Clinical Trials, and Seamless Dyadic Interaction

EXPERIENCE

Research Assistant, *Signal Analysis and Interpretation Laboratory (SAIL)*

Sep 2023-Present

- Develop explainable multimodal models of human temporal dynamics across video and speech, with 10+ publications
- Partner with 3+ multidisciplinary teams (law enforcement, healthcare, linguistics, psychology) to build AI pipelines

PROJECTS

Speech Production on Real-time MRI Video

Sep 2023-Present

- Build a video world model for rtMRI speech production analysis, Arti-JEPA, that doubling cross-domain phoneme decoding over pretrained baselines and offering various applications to clinical speech (CSL Submission)
- Developed the first audio-guided diffusion model, Speech2rtMRI, for real-time MRI synthesis of vocal tract, with up to 34% of samples misidentified as authentic by expert phoneticians (ICASSP 2025)
- Engineered multimodal video encoder for real-time MRI by fusing optical flow with 6-channel region-of-interest inputs that reduce phoneme recognition error rate by 5% (ICASSP 2026)

Temporal Dynamics in Human Speech and Gestures

Oct 2024-Present

- Developed MOOSE, a motion-centric video encoder for fine-grained human activity understanding (TMLR submission)
- Improved detection of atypical motor behaviors in autism by 6% accuracy, with enhanced saliency-based explainability
- Audited evaluation practices in co-speech gesture research, showing learned judges exploit speaker and session shortcuts (0.92 accuracy with no true coupling), and proposed 3 controls for rigorous evaluation (ICASSP 2027 submission)

Egocentric Officer-Civilian Interaction on Traffic Stops

Dec 2024-Present

- Design Video Embedding Extraction pipeline and Setting Selection on 30K+ body-worn videos of traffic stops
- Analyze 2K+ Annotated Videos of Multimodal Collaborative Officer-Civilian Activity from Multiple Views

Multimodal Human Trust Perception

Sep 2025-Mar 2026

- Partner with Capital One to collect and benchmark EVALUATE dataset (920 ads), exposing gaps in Emotion and Trust
- Showed via label-stability analysis that survey design systematically shifts human annotator judgments

Image Severity Ranking in Ophthalmology

Sep 2022-Dec 2023

- Designed ConPrO, a severity representation learning framework combining contrastive learning and preference optimization, outperforming state-of-the-art methods by 6% in F1 score (CVPRW 2024)
- Developed SiameN and SEMISE, expert-centric severity ranking models, improving IoU explainability by 6% and ranking performance by 3% in terms of F1 score (ISBI 2025, EMBC 2025 and OpthSci 2025)

TECHNICAL SKILLS

- **Programming Language:** Python, C/C++, Linux / Shell Scripting
- **Tools:** Coding Agent (Claude Code, Codex), Distributed Computing (HPC, Kubernetes, AWS, GCP), Machine Learning (Pytorch/XLA, Huggingface, OpenCV, Scikit-learn, Matplotlib), Package Management (Git, Conda)

LEADERSHIP AND INVOLVEMENT

- **Activities:** Research Mentoring (5+ Mentee), CVPR/NeurIPS Reviewer (5+ Papers), Proposals Writing (2), Webpage Management (2), Temasek Community Action and Leadership Exchange (2019), USC ECE Research Festival (2024)
- **Awards:** VinGroup Fellowship, 1st-place at 2022 Zalo AI Challenge, Honda Young Engineers and Scientists Award